4.6.3 Manually Set the Tool Offset

To manually set tool offsets:

1. Choose one of the tool offsets pages.
2. Move the cursor to the desired column.
3. Type a number and press **[ENTER]** or **[F1]**.

Pressing **[F1]** enters the number in the selected column. Entering a value and pressing **[ENTER]** adds the amount entered to the number in the selected column.

4.6.4 Setting Part Zero for the Z-axis (Part Face)

Your CNC control programs all move from Part Zero, a user-defined reference point. To set Part Zero:

1. Press **[MDI/DNC]** to select Tool #1.
2. Enter `T1` and press **[TURRET FWD]**.
3. Jog X and Z until the tool just touches the face of the part.
4. Press **[OFFSET]** until the **`Work Zero Offset`** display is active. Highlight the **`Z Axis`** column and G-code row that you want to use (G54 recommended).
5. Press **[Z FACE MEASURE]** to set part zero.

4.7 Chuck and Collet Replacement

These procedures describe how to remove and replace a chuck or collet.

For detailed instructions on the procedures listed in this section, go to www.HaasCNC.com and select Owners> WORK ON YOUR HAAS.

4.7.1 Chuck Installation

To install a chuck:

NOTE: *If necessary, install an adapter plate before installing the chuck.*

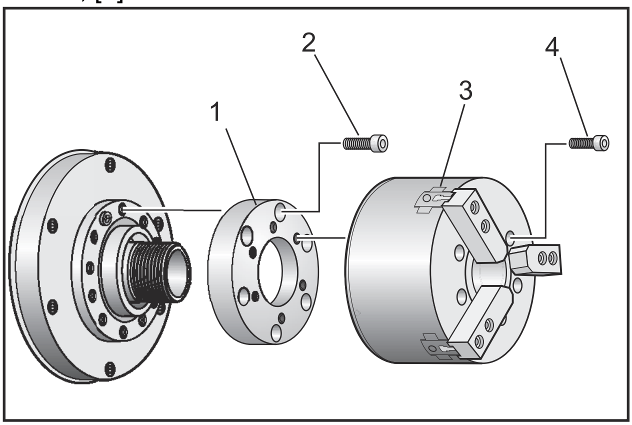
1. Clean the face of the spindle and the back face of the chuck. Position the drive dog at the top of the spindle.
2. Remove the jaws from the chuck. Remove the center cup or coverplate from the front of the chuck. If available, install a mounting guide into the drawtube and slide the chuck over it.

3. Orient the chuck so that one of the guide holes are aligned with the drive dog. Use the chuck wrench to thread the chuck onto the drawtube.
4. Screw the chuck all the way onto the drawtube and back it off 1/4 turn. Align the drive dog with one of the holes in the chuck. Tighten the six (6) SHCS.
5. Install the center cup or plate with three (3) SHCS.
6. Install the jaws. If necessary replace the rear cover plate. This is located on the left side of the machine.

4.7.2 Chuck Removal

This is a summary of the chuck removal process.

F4.11: Chuck Removal Illustration: [1] Chuck Adapter Plate, [2] 6X Socket Head Cap Screws (SHCS), [3] Chuck, [4] 6X SHCS.



1. Move both axes to their zero positions. Remove the chuck jaws.
2. Remove the three (3) screws that mount the center cup (or plate) from the center of the chuck and remove the cup.

CAUTION: *You must clamp the chuck when you do this next step, or you will damage the drawtube threads.*

3. Clamp the chuck [3] and remove the (6) SHCS [4] that mount the chuck to the spindle nose or adapter plate.
4. Unclamp the chuck. Place a chuck wrench inside the center bore of the chuck and unscrew the chuck from the drawtube. If equipped, remove adapter plate [1].

WARNING: *The chuck is heavy. Be prepared to use lifting equipment to support the chuck while you remove it.*

4.7.3 Chuck/Drawtube Warnings

WARNING: *Check the workpiece in the chuck or collet after any power loss. A power outage reduces the clamping pressure on the workpiece which can shift in the chuck or collet. Setting 216 turns off the Hydraulic pump after the time specified for the setting.*

WARNING: *Damage results if you attach dead length stops to the hydraulic cylinder.*

WARNING: *Do not machine parts larger than the chuck.*

WARNING: *Follow all of the warnings of the chuck manufacturer.*

WARNING: *Hydraulic pressure must be set correctly. See the* **`Hydraulic System Information`** *on the machine for safe operation. Setting a pressure beyond the recommendations damages the machine and/or inadequately holds the workpiece.*

WARNING: *Chuck jaws must not protrude beyond the diameter of the chuck.*

WARNING: *Improperly or inadequately clamped parts eject with deadly force.*

WARNING: *Do not exceed rated chuck RPM.*

WARNING: *Higher RPM reduces chuck clamping force. Refer to the chart.*

NOTE: *Grease your chuck weekly, and keep it clean.*

4.7.4 Collet Installation

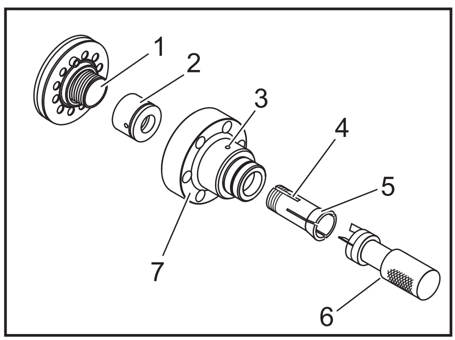
To install a collet:

1. Thread the collet adapter into the drawtube.
2. Place the spindle nose on the spindle and align one of the holes on the back of the spindle nose with the drive dog.
3. Fasten the spindle nose to the spindle with six (6) SHCS.
4. Thread the collet onto the spindle nose and align the slot on the collet with the set screw on the spindle nose. Tighten the setscrew on the side of the spindle nose.

4.7.5 Collet Removal

To remove the collet:

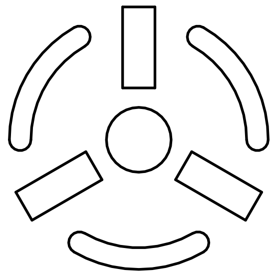
F4.12: Collet Removal Illustration: [1] Draw Tube, [2] Collet adapter, [3] Set Screw, [4] Set Screw Slot, [5] Collet, [6] Collet wrench, [7] Spindle Nose.



1. Loosen the set screw [3] on the side of the spindle nose [7]. Using the collet wrench [6], unscrew the collet [5] from the spindle nose [7].
2. Remove the six (6) SHCS from the spindle nose [7] and remove it.
3. Remove the collet adapter [2] from the drawtube [1].

4.7.6 Chuck Foot Pedal

F4.13: Chuck Foot Pedal Icon



NOTE: *Dual-spindle lathes have a pedal for each chuck. The relative positions of the pedals indicate the chuck that they control (i.e., the left-hand pedal controls the main spindle and the right-hand pedal controls the secondary spindle).*

When you press this pedal, the automatic chuck clamps or unclamps, equivalent to an `M10` / `M11` command for the main spindle, or `M110` / `M111` command for the secondary spindle. This allows you to operate the spindle hands-free while you load or unload a workpiece.

The ID / OD clamp settings for the main and secondary spindles apply when you use this pedal (refer to Setting 282 on page **400** for more information).

Use Setting 332 to enable or disable all pedal controls. Refer to Setting 332 on page **403**

4.7.7 Steady Rest Foot Pedal

F4.14: Steady Rest Foot Pedal Icon